

Ms. Lipi Kasera

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Portfolio: <https://lipikasera.github.io>

SUMMARY

GIS and Spatial Data Analyst with a Master of GIS from the University of Queensland and a proven ability to transform complex spatial data into clear, actionable insights. Proficient in ArcGIS Pro, QGIS, Python (GeoPandas) and SQL, with hands-on experience delivering end-to-end geospatial projects across spatial analysis, environmental modelling and geospatial data management. Successfully processed and classified 10,000+ geological records to produce high-quality spatial datasets and maps and developed coastal hazard and erosion risk models using advanced forecasting techniques, including closure depth analysis, the Bruun Rule and Monte Carlo simulations. Recognised for combining strong technical expertise, analytical thinking and attention to detail with the ability to communicate complex findings effectively to both technical and non-technical audiences. Passionate about applying geospatial technologies, data-driven decision-making and reproducible analytical workflows to solve real-world environmental, infrastructure and spatial challenges, and ready to contribute meaningfully to graduate GIS, environmental or spatial data roles.

EDUCATION

- **Master of Geographical Information Systems, University of Queensland, 2025**
Coursework: Spatial Analysis, Geospatial Modelling, Environmental GIS, Spatial Databases, Python for GIS, Urban Planning & Design. Skills: GIS data management, infrastructure planning, land use analysis, built form assessment, documentation and urban growth modelling.
- **Bachelor of Arts (Geography), Miranda House, University of Delhi, 2023**

TECHNICAL SKILLS

- **GIS Platforms:** ArcGIS Pro, ArcGIS Online, QGIS, ENVI
- **Programming:** Python (Geopandas, NumPy, Pandas), SQL, ModelBuilder
- **Spatial Analysis:** Geospatial modelling, interpolation, terrain analysis, coastal modelling, demographic & land use analysis
- **Data Management:** Spatial data integration, database management, data transformation, data validation & QA
- **Visualisation:** Cartographic design, spatial data visualisation, ArcGIS Online dashboards, Adobe Illustrator, SketchUp
- **Field Methods:** Dumpy level surveying, GPS data collection, topographic surveys, field data validation
- **Professional Attributes:** Strong attention to spatial data accuracy and analytical rigour; Fast learner with a growth mindset, comfortable adapting to new tools, workflows and technical environments; Effective collaborator with experience supporting senior researchers and contributing to team deliverables under deadline; Clear written and verbal communicator; Experienced in translating complex spatial outputs into accessible reports for non-technical audiences.

CAREER PROJECTS

Geological Mapping & Rock Age Analysis | University of Queensland | December 2024 – May 2025

Overview: Spatial analysis project mapping rock formation distribution and age across Queensland using real geological survey datasets.

Responsibilities

- Processed and standardised 10,000+ geological records in MS Excel and ArcGIS Pro, applying QA checks to identify and resolve data inconsistencies before spatial analysis
- Produced 8–10 geological maps using GIS-based classification, interpolation and visualisation techniques to identify rock formation patterns and spatial data gaps across Queensland
- Prepared a technical report translating spatial findings into targeted recommendations for future survey sampling locations

Flinders Beach Coastal Hazard Assessment | University of Queensland | February 2025 – May 2025

Overview: Evidence-based coastal hazard assessment integrating wave climate data, beach profile surveys and probabilistic modelling to project shoreline change under future scenarios.

Responsibilities

- Analysed beach profile and wave climate datasets to characterise shoreline dynamics and identify erosion trends at Flinders Beach
- Applied coastal modelling techniques — including closure depth analysis, the Bruun Rule and Monte Carlo simulations — to generate probabilistic shoreline change projections across multiple risk scenarios
- Delivered a comprehensive hazard report identifying high-risk erosion zones and infrastructure exposure, with uncertainty quantification to support evidence-based coastal management recommendations

References are available on request.